



University
of Glasgow

MSc Project Topics

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**WORLD
CHANGING
GLASGOW**

THE SUNDAY TIMES
GOOD
UNIVERSITY
GUIDE
2022

SCOTTISH
UNIVERSITY
OF THE YEAR

A bit about myself



Hanoi University of Science and Technology, Vietnam (BSc)



Pohang University of Science and Technology, South Korea (MSc)



DASAN Networks Corporation, South Korea (Software Engineer)



Liverpool John Moores University, Ph.D (2015-2018)



Imperial College London, Research Associate (2018-2022)



University of Glasgow, Lecturer (2022-onward)

MSc Research Projects

1. **Private Browsers:**

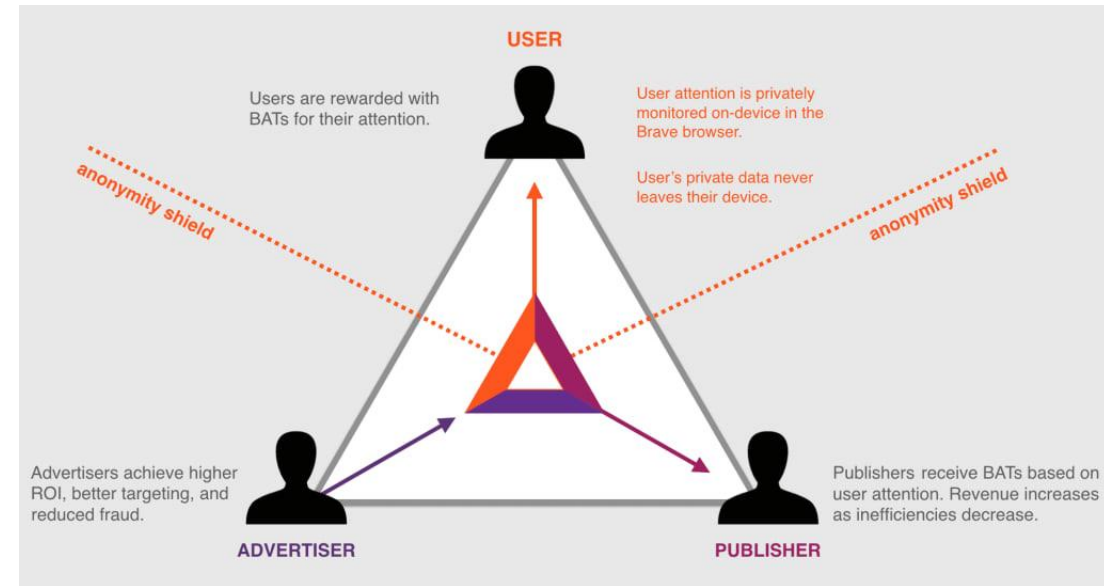
- Incognito mode: limitations; Privacy Extension: ad-blocking extension
- Brave: a user-friendly privacy-preservation private browser

Brave:

- Reward Scheme: Brave leverages Blockchain technology for incentivizing users
- BAT tokens

Privacy in Brave?

- Re-identify Brave users using blockchain-based de-anonymity techniques



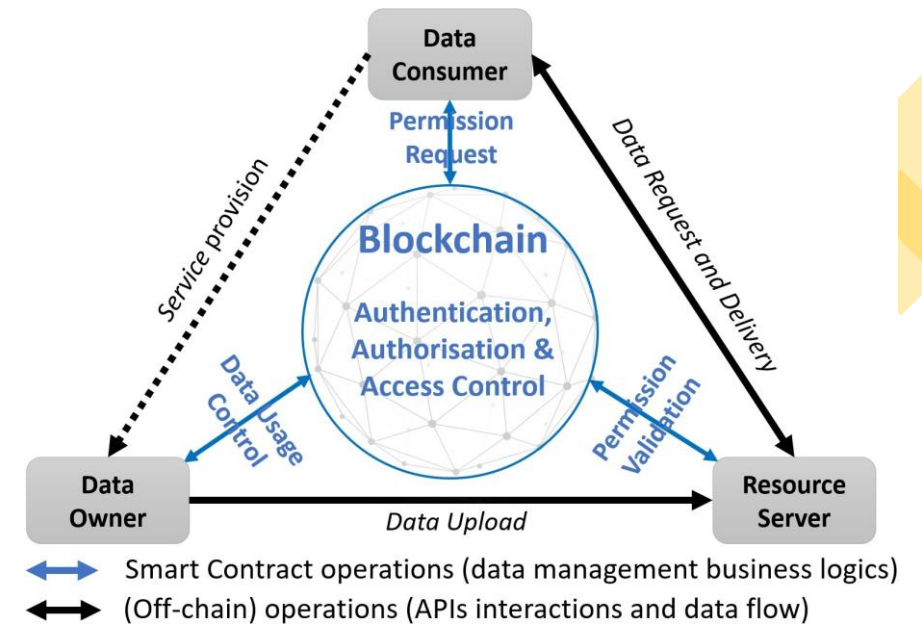
Some knowledge/skill required to accomplish:

1. Analytical skill
2. A bit about graph theory
3. A bit about Bitcoin/Blockchain
4. Any programming languages

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2. Decentralised Data Management:

- Your data is stored in any storage systems
- You fully control your data, can sell, share, or prevent any services from accessing it.
- Leverage Blockchain technology with advanced cryptography techniques
 - Encryption schemes
 - Proxy Encryptions
 - Access Tokens



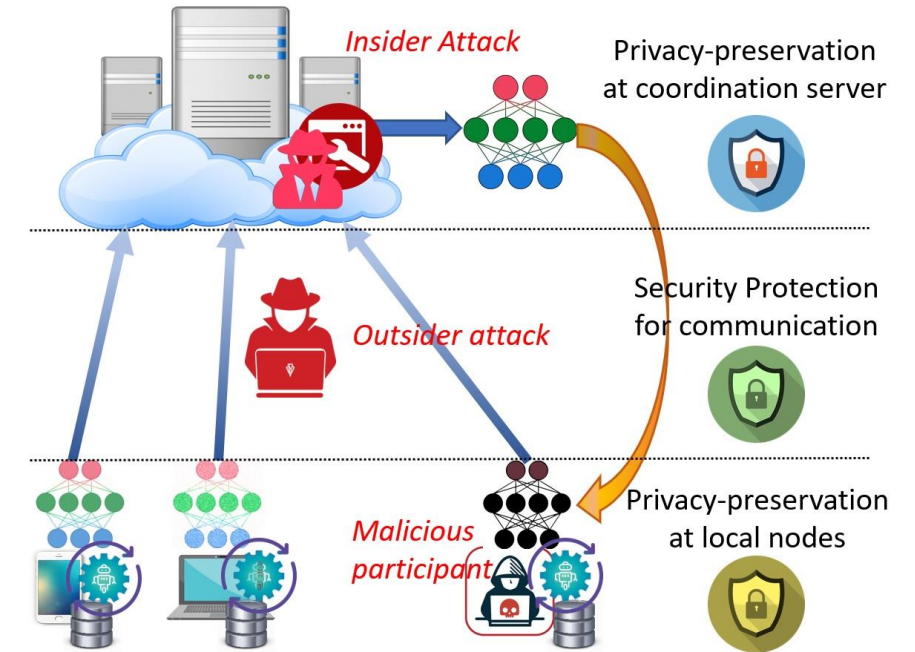
Some knowledge/skill required to accomplish:

1. Understanding of Bitcoin/Blockchain/ and Ethereum
2. Public-key Cryptography
3. Programming Languages:
 1. Web app development (any)
 2. Solidity (don't worry, it's simple -> grasp it quickly)

MSc Research Projects

3. Privacy in Federated Learning:

- Train ML models in a decentralized collaborative learning settings
- Trained data is stored in isolated sources
- Federated Learning technique retains data and computation at local sites, only results of the (local) training are exchanged (aggregated and updated).
- Privacy-preservation techniques for the parameters' aggregation and updates.
- Possible attacks and solutions



Some knowledge/skill required to accomplish:

1. Analytical Skill
2. Basic Machine Learning/Deep Learning
3. Cryptography including Multiparty computation (MPC)
4. Programming Languages:
 1. Python (the most suitable for ML tasks)
 2. Otherwise, any languages



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Thank you for
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